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Paper Code : BSCAIDS301/BSCAIML 301/BSCAIML301/BSCICB301 Linear Algebra

UPID : 003902

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

## Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[ 1 x 10 = 10 ]

- (i) If A and B are orthogonal matrices of the same order then AB is \_\_\_\_\_.
- (ii) Let V be of finite dimension and let  $F: V \rightarrow U$  be a linear mapping. Then  $\dim(\text{Ker } F) + \dim(\text{Im } F) = \underline{\hspace{2cm}}$ .
- (iii) An orthonormal set can always be obtained from an orthogonal set of nonzero vectors by normalizing each vector (a) True (b) False.
- (iv) The norm of  $v = (3, 4)$  with respect to the usual inner product is \_\_\_\_\_.
- (v) The inverse of a square matrix is unique. (a) True (b) False.
- (vi) If A, B, C are three square matrices such that  $AB = BC = I$ , then  $A = ?$
- (vii) The dimension or rank of a finite dimensional vector space V is the number of vectors in any basis for V. (a) True (b) False.
- (viii) In the Euclidean space let  $u = (-2, -1, 0)$  and  $v = (3, -6, 1)$  then check whether u and v are orthogonal vectors or not.
- (ix) According to Schur's unitary triangularization theorem, every matrix is unitarily equivalent to a \_\_\_\_\_.
- (x) If A is a symmetric as well as a skew symmetric matrix then A must be a \_\_\_\_\_ matrix.
- (xi) If  $S = M + iN$  be a Skew Hermitian matrix, then the diagonal elements of S are all \_\_\_\_\_.
- (xii) Is the mapping  $F(x, y) = (x+1, 2y, x+y)$  linear? Justify.

## Group-B (Short Answer Type Question)

Answer any three of the following :

[ 5 x 3 = 15 ]

2. Find the matrix of the linear transformation on  $V^3(\mathbb{R})$  defined as  $T(a, b, c) = (2b + c, a - 4b, 3a)$  with respect to the standard basis  $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ . [5]
3. Find the norm of  $\alpha = (-1, 2, 3)$  with standard inner product. Normalize the vector. [5]
4. Find the matrix of the linear transformation on  $V^3(\mathbb{R})$  defined as  $T(a, b, c) = (2b + c, a - 4b, 3a)$  with respect to the ordered basis  $B = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$ . [5]
5. Examine whether the mapping  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  defined by  $T(x, y) = (2x - y, x)$  is linear. [5]
6. Let T be a linear operator on  $\mathbb{R}^2$  defined by  $T(3, 1) = (2, -4)$  and  $T(1, 1) = (0, 2)$ . Find  $T(a, b)$ . [5]

## Group-C (Long Answer Type Question)

Answer any three of the following :

[ 15 x 3 = 45 ]

7. (a) Express the vector  $(-1, 2, 4)$  as a linear combination of the vectors  $(-1, 2, 0)$ ,  $(0, -1, 1)$  and  $(3, -4, 5)$  in the vector space  $\mathbb{R}^3$  over the real field. [5]
- (b) Let S and T be linear operators on  $\mathbb{R}^2$  defined by  $S(x, y) = (0, x)$  and  $T(x, y) = (x, 0)$ . Show that  $TS = 0$  but  $ST \neq 0$ . Also show that  $T^2 = T$ . [5]
- (c) Let  $T: V \rightarrow W$  be a linear transformation. Prove that  $T(\alpha) = T(\beta) \Rightarrow \alpha = \beta$  if  $\ker(T) = \{0\}$ . [5]
8. (a) Determine the matrix for the linear transformation  $T: \mathbb{R}^5 \rightarrow \mathbb{R}^3$  defined by  $T(x, y, z, p, q) = (y, z, p)$  with respect to the standard basis. [5]
- (b) The linear transformation  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$  is defined by  $T(x, y, z) = (2x + y - z, y + 4z, x - y + 3z)$ . Find the matrix of T relative to the ordered basis  $\{(0, 1, 1), (1, 0, 1), (1, 1, 0)\}$ . Hence find  $\text{rank}(T)$  and state whether  $T^{-1}$  exists. [5]
- (c) If  $T_1$  and  $T_2$  be two linear transformations on a vector space  $V(F)$  such that  $T_1 T_2 = T_2 T_1$ . then prove that  $(T_1 + T_2)^2 = (T_1)^2 + 2 T_1 T_2 + (T_2)^2$ . [5]

9. (a) In  $\mathbb{R}^3$  let  $\alpha = (x_1, x_2, x_3)$ ,  $\beta = (y_1, y_2, y_3)$  then find whether  $\alpha, \beta$  defined as  $(x_1+x_2+x_3)(y_1+y_2+y_3)$  is a real inner product. [7]
- (b) Extend the orthogonal set  $A = \{(3, 0, 4), (-4, 0, 3)\}$  to an orthogonal basis of  $\mathbb{R}^3$  and find the associated orthonormal basis. [8]
10. (a) Prove that the set of vectors  $\{(1, 2, -2), (2, 1, 2), (2, -2, -1)\}$  is an orthogonal set of vectors of the inner product space  $\mathbb{R}^3$  with standard inner product. Convert this orthogonal set to an orthonormal set. Express  $(1, 0, 2)$  as a linear combination of the vectors of the given set. [8]
- (b) Apply Gram – Schmidt process to the vectors  $b_1 = (1, 0, 1)$ ,  $b_2 = (1, 0, -1)$ ,  $b_3 = (0, 3, 4)$  to obtain an orthonormal basis for  $V_3(\mathbb{R})$  with the standard inner product. [7]
11. (a) Using LU decomposition method, solve the following:  $x + 2y + 4z = 3$ ;  $3x + 8y + 14z = 13$ ;  $2x + 6y + 13z = 4$ . [8]
- (b) Prove that every square matrix can be uniquely expressed as the sum of a symmetric and a skew symmetric matrix. [7]

\*\*\* END OF PAPER \*\*\*